P2 Chapter 8: Parametric Equations

Parametric Modelling

Modelling

As we saw at the start of this chapter, parametric equations are frequently used in mechanics, particularly where the (x, y) position (the Cartesian variables) depends on time t (the parameter).

[Textbook] A plane's position at time t seconds after take-off can be modelled with the following parametric equations:

$$x = (v\cos\theta)t \text{ m}, \qquad y = (v\sin\theta)t \text{ m}, \qquad t > 0$$

where v is the speed of the plane, θ is the angle of elevation of its path, x is the horizontal distance travelled and y is the vertical distance travelled, relative to a fixed origin.

When the plane has travelled 600m horizontally, it has climbed 120m.

a. find the angle of elevation, θ.

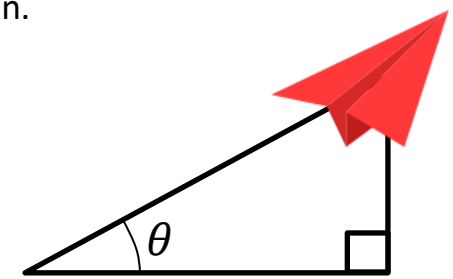
Given that the plane's speed is 50 m s^{-1},

b. find the parametric equations for the plane's motion.

c. find the vertical height of the plane after 10 seconds.

d. show that the plane's motion is a straight line.

e. explain why the domain of t, $t > 0$, is not realistic.



a ?

b ?

c ?

d ?

e ?

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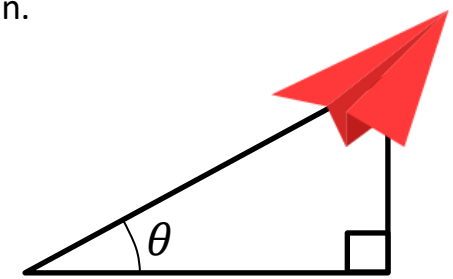


a $\theta = \tan^{-1}\left(\frac{120}{600}\right) = 11.3° \ (1dp)$

b $x = (v\cos\theta)t$

$= (50 \times \cos 11.3\ldots)t = 49.0t \text{ m } (3sf)$

$y = (v\sin\theta)t = (50 \times \sin 11.3\ldots) = 9.80t \text{ m } (3sf)$

c At $t = 10$, $y = 9.80t = 98$ m

d $x = 49t \quad \rightarrow \quad t = \frac{x}{49}$ and $y = 9.8 \times \frac{x}{49} = \frac{1}{5}x$

$\therefore y = \frac{1}{5}x$ which is a linear equation.

e Plane would continue to climb forever!

Further Example

[Textbook] The motion of a figure skater relative to a fixed origin, O, at time t minutes is modelled using the parametric equations

$$x = 8\cos 20t\,, \qquad y = 12\sin\left(10t - \frac{\pi}{3}\right), \qquad t \geq 0$$

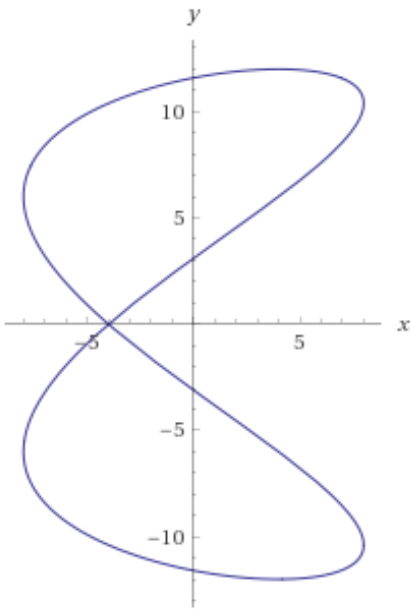
where x and y are measured in metres.

a) Find the coordinates of the figure skater at the beginning of his motion.

b) Find the coordinates of the point where the figure skater intersects his own path.

c) Find the coordinates of the points where the path of the figure skater crosses the y-axis.

d) Determine how long it takes the figure skater to complete one complete figure-of-eight motion.



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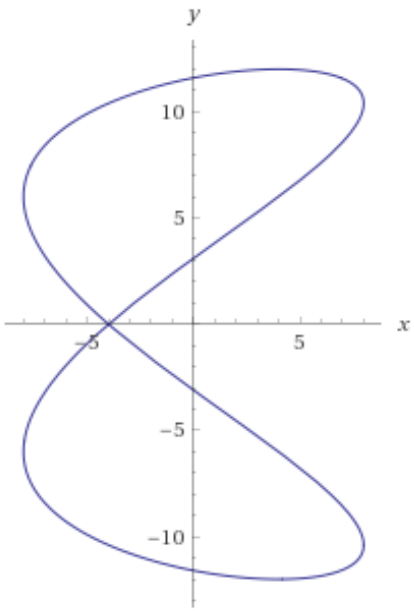

a At $t = 0$,

$$x = 8\cos 0 = 8, \qquad y = 12\sin\left(-\frac{\pi}{3}\right) = -6\sqrt{3}$$

b From the graph, the intersection is when $y = 0$:

$$12\sin\left(10t - \frac{\pi}{3}\right) = 0$$

$$\ldots$$

$$t = \frac{\pi}{30}, \frac{2\pi}{15}, \frac{7\pi}{30}, \ldots$$

(Any of these values will do)

$$x = 8\cos\left(20 \times \frac{\pi}{30}\right) = -4 \quad \rightarrow \quad (-4,0)$$

c When $x = 0$:

$$0 = 8\cos t$$

$$t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots$$

Substitute these t-values into y.

$$t = \frac{\pi}{2} \rightarrow \quad y = 3.11\ (2dp)$$

$$t = \frac{3\pi}{2} \rightarrow \quad y = 11.59 \quad \ldots$$

$$(0, -3.11), (0,11.59), (0,3.11), (0, -11.59)$$

d The period of $\cos t$ is 2π, so the period of $\cos 20t$ is $\frac{2\pi}{20}$ (as graph stretched by factor of $\frac{1}{20}$). Similarly period of y is $\frac{2\pi}{10}$. Both parametric equations must complete full periods (in this case so that skater returns to original y position), so period is $\frac{2\pi}{10}$ min = 38 seconds.

Exercise 8.5

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Homework Exercise

1 A river flows from north to south. The position at time t seconds of a rowing boat crossing the river from west to east is modelled by the parametric equations

$$x = 0.9t \text{ m}, \quad y = -3.2t \text{ m}$$

where x is the distance travelled east and y is the distance travelled north.

Given that the river is 75 m wide,

a find the time taken to get to the other side

b find the distance the boat has been moved off-course due to the current

c show that the motion of the boat is a straight line

d determine the speed of the boat.

2 The position of a small plane coming into land at time t minutes after it has started its descent is modelled by the parametric equations

$$x = 80t, \quad y = -9.1t + 3000, \quad 0 \leqslant t < 330$$

where x is the horizontal distance travelled (in metres) and y is the vertical distance travelled (in metres) from the point of starting its descent.

a Find the initial height of the plane.

b Justify the choice of domain, $0 \leqslant t < 330$, for this model.

c Find the horizontal distance the plane travels between beginning its descent and landing.

3 A ball is kicked from the ground with an initial speed of 20 m s^{-1} at an angle of 30°. Its position after t seconds can be described using the following parametric equations

$$x = 10\sqrt{3}\,t \text{ m}, \quad y = (-4.9t^2 + 10t) \text{ m}, \quad 0 \leqslant t \leqslant k$$

a Find the horizontal distance travelled by the ball when it hits the ground.

A player wants to head the ball when it is descending between 1.5 m and 2.5 m off the ground.

b Find the range of time after the ball has been kicked at which the player can head the ball.

c Find the closest distance from where the ball has been kicked at which the player can head the ball.

Homework Exercise

4 The path of a dolphin leaping out of the water can be modelled with the following parametric equations

$$x = 2t \text{ m}, \quad y = -4.9t^2 + 10t \text{ m}$$

where x is the horizontal distance from the point the dolphin jumps out of the water, y is the height above sea level of the dolphin and t is the time in seconds after the dolphin has started its jump.

a Find the time the dolphin takes to complete a single jump.

b Find the horizontal distance the dolphin travels during a single jump.

c Show that the dolphin's path is modelled by a quadratic curve.

d Find the maximum height of the dolphin.

5 The path of a car on a Ferris wheel at time t minutes is modelled using the parametric equations

$$x = 12\sin t, \quad y = 12 - 12\cos t$$

where x is the horizontal distance in metres of the car from the start of the ride and y is the height in metres above ground level of the car.

a Show that the motion of the car is a circle with radius 12 m.

b Hence, find the maximum height of the car during the journey.

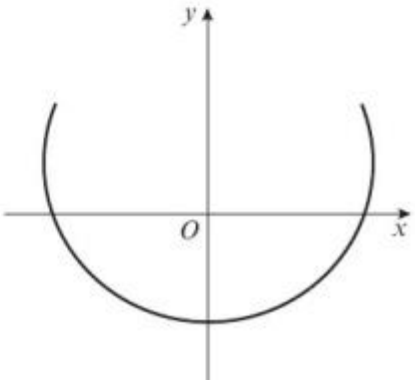
c Find the time taken to complete one revolution of the Ferris wheel and hence calculate the average speed of the car.

6 The cross-section of a bowl design is given by the following parametric equations

$$x = t - 4\sin t, \quad y = 1 - 2\cos t, \quad -\frac{\pi}{2} \leqslant t \leqslant \frac{\pi}{2}$$

a Find the length of the opening of the bowl. **(3 marks)**

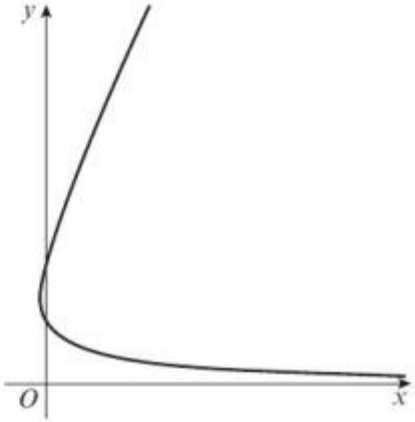
b Given that the cross-section of the bowl crosses the y-axis at its deepest point, find the depth of the bowl. **(4 marks)**



Homework Exercise

7 A particle is moving in the xy-plane such that its position after time t seconds relative to a fixed origin O is given by the parametric equations

$$x = \frac{t^2 - 3t + 2}{t}, \quad y = 2t, \quad t > 0$$



The diagram shows the path of the particle.

a Find the distance from the origin to the particle at time $t = 0.5$.

b Find the coordinates of the points where the particle crosses the y-axis.

Another particle travels in the same plane with its path given by the equation $y = 2x + 10$.

c Show that the paths of these two particles never intersect.

8 The path of a ski jumper from the point of leaving the ramp to the point of landing is modelled using the parametric equations

$$x = 18t, \quad y = -4.9t^2 + 4t + 10, \quad 0 \leqslant t \leqslant k$$

where x is the horizontal distance in metres from the point of leaving the ramp and y is the height in metres above ground level of the ski jumper, after t seconds.

a Find the initial height of the ski jumper. **(1 mark)**

b Find the value of k and hence state the time taken for the ski jumper to complete her jump. **(3 marks)**

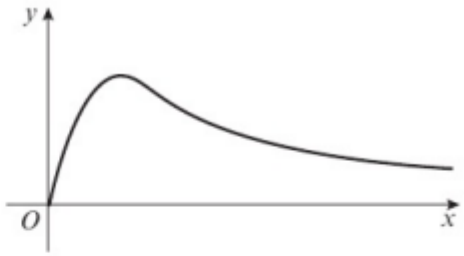
c Find the horizontal distance the ski jumper jumps. **(1 mark)**

d Show that the ski jumper's path is a parabola and find the maximum height above ground level of the ski jumper. **(5 marks)**

Homework Exercise

9 The profile of a hill climb in a bike race is modelled by the following parametric equations

$$x = 50\tan t\,\text{m}, \quad y = 20\sin 2t\,\text{m}, \quad 0 < t \leqslant \frac{\pi}{2}$$



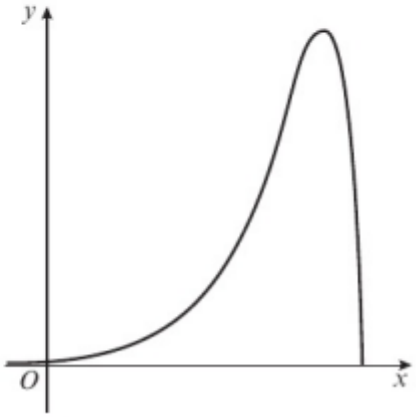
a Find the value of t at the highest point of the hill climb.

b Hence find the coordinates of the highest point.

c Find the coordinates when $t = 1$ and show that at this point, a cyclist will be descending.

10 A computer model for the shape of the path of a rollercoaster is given by the parametric equations

$$x = 5 + \ln t, \quad y = 5\sin 2t, \quad 0 < t \leqslant \frac{\pi}{2}$$



a Find the coordinates of the point where $t = \frac{\pi}{6}$ **(2 marks)**

Given that one unit on the model represents 5 m in real life,

b find the maximum height of the rollercoaster **(1 mark)**

c find the horizontal distance covered during the descent of the rollercoaster. **(4 marks)**

d Hence, find the average gradient of the descent. **(1 mark)**

Homework Answers

1 **a** 83.3 seconds **b** 267 m

c $t = \frac{x}{0.9} \Rightarrow y = -3.2\frac{x}{0.9} \Rightarrow y = -\frac{32}{9}x$

which is in the form, $y = mx + c$ and is therefore a straight line.

d $3.32\,\text{ms}^{-1}$

2 **a** 3000 m

b Initial point is when $t = 0$. For $t \geqslant 330$, y is negative ie. the plane is underground or below sea level.

c 26 400 m (3 s.f.)

3 **a** 35.3 m

b Between 1.75 and 1.88 seconds (3 s.f.)

c 30.3 (3 s.f.).

4 **a** $\frac{100}{49}$ seconds

b $\frac{200}{49}$ m

c $t = \frac{x}{2} \Rightarrow y = -4.9\left(\frac{x}{2}\right)^2 + 10\left(\frac{x}{2}\right) = -\frac{49}{40}x^2 + 5x$

Therefore, the dolphin's path is a quadratic curve

d $\frac{250}{49}$ m

5 **a** $\sin t = \frac{x}{12}$, $\cos t = \frac{y-12}{-12}$

$\left(\frac{x}{12}\right)^2 + \left(\frac{y-12}{-12}\right)^2 = 1 \Rightarrow x^2 + (y-12)^2 = 144$

Therefore, motion is a circle with centre (0,12) and radius $\sqrt{144} = 12$.

b 24 m **c** 2π minutes, 12 m/min

6 **a** 4.86 (3 s.f.) **b** Depth = 2

7 **a** $\frac{\sqrt{13}}{2}$ **b** (0, 2), (0, 4)

c $2t = 2\left(\frac{t^2 - 3t + 2}{t}\right) + 10$

$2t^2 = 2t^2 - 6t + 4 + 10t \Rightarrow 0 = 4t + 4 \Rightarrow t = -1$

Since, $t > 0$, the paths do not intersect.

8 **a** 10 m

b $k = 1.89$ (3 s.f.). Therefore, time taken is 1.89 seconds.

c 34.1 m (3 s.f.)

d $t = \frac{x}{18}$

$y = -4.9\left(\frac{x}{18}\right)^2 + 4\left(\frac{x}{18}\right) + 10 = -\frac{49}{3240}x^2 + \frac{2}{9}x + 10$

Therefore, the ski jumper's path is a quadratic equation. Maximum height = 10.8 m (3 s.f.)

9 **a** $t = \frac{\pi}{4}$ **b** (50, 20)

c (77.87, 18.19)

$\frac{\pi}{4} < 1 < \frac{\pi}{2}$, which is when $\sin 2t$ is decreasing, hence when y is decreasing, hence the cyclist is descending.

10 **a** (4.35, 4.33) (3 s.f.) **b** 25 m

c 3.47 m (3 s.f.) **d** −7.21